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by muscle pull due to improper posture with static sitting for long hours.

Muscular dystrophy: The group of diseases collectively called the muscular dystrophy are associated with the progressive degeneration of skeletal muscle fibres, weakening the muscles and leading to death from lung or heart failure. The most common form of muscular dystrophy is called Duchene Muscular Dystrophy (DMD).

b) Disorders of skeletal system

Arthritis and osteoporosis are the major disorders of skeletal system.

1. **Arthritis:** Arthritis is an inflammatory (or) degenerative disease that damages the joints. There are several types of arthritis.
   - **(i) Osteoarthritis:** The bone ends of the knees and other freely movable joints wear away as a person ages. The joints of knees, hip, fingers and vertebral column are affected.
   - **(ii) Rheumatoid arthritis:** The synovial membranes become inflamed and there is an accumulation of fluid in the joints. The joints swell and become extremely painful. It can begin at any age but symptoms usually emerge before the age of fifty.
   - **(iii) Gouty arthritis or gout:** Inflammation of joints due to accumulation of uric acid crystals or inability to excrete it. It gets deposited in synovial joints.
2. **Osteoporosis:** It occurs due to deficiency of vitamin D and hormonal imbalance. The bone becomes soft and fragile. It causes rickets in children and osteomalacia in adult females. It can be minimized with adequate calcium intake, vitamin D intake and regular physical activities.

DTI – Diffusion Tensor Imaging is applied to study skeletal muscle physiology, anatomy and pathology.

9.13 Benefits of regular Exercise

Exercise and physical activity fall into four basic categories. Endurance, strength, balance and flexibility.

Endurance or aerobic activities increase the breathing and heart rate. They keep the circulatory system healthy and improve overall fitness.

Strength exercises make the muscles stronger. They help to stay independent and carry out everyday activities such as climbing stairs and carrying bags.

Balance exercises help to prevent falls which is a common problem in older adults. Many strengthening exercises also improves balance.

Flexibility exercises help to stretch body muscles for more freedom of joint movements.

Regular exercises can produce the following beneficial physiological changes:

- The muscles used in exercise grow larger and stronger.
- The resting heart rate goes down.
- More enzymes are synthesized in the muscle fibre.
- Ligaments and tendons become stronger.